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COURSE NAME: DATA SCIENCE

BRIGHTLEARN EXERCISE 5 NULL FUNCTIONS

1. SELECT employees, id,
name,
IFNULL (salary, 0) AS salary-with-default
FROM employees;

employee-id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

~~2. SELECT order-id,~~
2. SELECT employee-id,
name,
IFNULL (department, 'Not Assigned') AS department-name
FROM employees;

employee-id	name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

3. SELECT order-id,
customer-id
FROM Orders
WHERE customer-id IS NULL;

Order-id	Customer-id
103	NULL
102	
101	

4. SELECT Order-id,
Customer-id,
IFNULL (delivery-date, 'Pending') AS delivery-status
FROM Orders;

Order-id	Customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

5. SELECT student-id,
name,
IFNULL (grade, 0) AS final-grade
FROM students;

Student-id	name	grade
1	Ethan	85
2	Maya	0
3	Olivia	90

6. SELECT COUNT(*) AS not-graded-count
FROM students
WHERE grade IS NULL;

not-graded-count 1

7. SELECT product-id,
name,
Price - IFNULL(discount, 0) AS final-price
FROM Products;

Product-id	name	final-price
501	key board	25
502	mouse	10
503	monitor	100

8. SELECT COUNT(*) AS missing-email-count
FROM customers
WHERE email IS NULL;

missing-email-count 2

9. SELECT customer-id,
name,
IFNULL(email, 'No Email') AS email-display
FROM customers;

customer-id	name	email-display
1	Linda	NO NO Email
2	Joseph	joseph@mail.com
3	Nita	NO Email

10. ~~SELECT~~ payment-id,
method

10. SELECT payment-id;
IFNULL (method, 'unknown') AS method-display-status
FROM Payments;

Payment-id	method-display-status
301	Credit
302	Paypal
303	Unknown.

11. SELECT item-id,
item-name,
IFNULL (quantity, 0) AS quantity-checked
FROM Inventory;

item-id	item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

12. SELECT emp-id,
COALESCE (bonus, 'commission') AS first-available-reward
FROM Employees-Extra;

emp-id	first-available-reward
1	300
2	100
3	NULL

13. SELECT COUNT(*) AS no-room-count
FROM classes
WHERE room IS NULL;

no-room-count
2

14. SELECT student-id,
date,
IFNULL(status, 'Not Marked') AS attendance-status
FROM Attendance;

student-id	date	attendance-status
1	2025-04-01	Not-Marked
2	2025-04-01	Present
3	2025-04-01	Absent

15. SELECT account-id,
IFNULL(account-type, 'Unknown') AS ~~display~~ type-display;
IFNULL(balance, 0) AS balance-checked
FROM Bank-Accounts;

account-id	type-display	balance-checked
A1	savings	0
A2	current	5000
A3	Unknown	2000

16. SELECT project-id,
title,
IFNULL(start-date, 'TBD') AS start-display
FROM Projects;

Project-id	title	Start-display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data migration	TBD

17. SELECT review-id,
product-id,
IFNULL(comment, 'No comment') AS comment-display,
IFNULL(rating, 0) AS rating-display
FROM Reviews;

review-id	Product-id	rating-display	rating
1	501	Great product	4
2	502	No comment	0
3	503	Works Fine	3

18. SELECT supplier-id,
name,
COALESCE(phone, alt_phone, 'NO contact') AS ^{contact-number} n
FROM Suppliers;

supplier-id	name	contact-number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	NO Contact

19. SELECT user-id,
IFNULL(theme, 'Light') AS theme-set,
IFNULL(language, 'English') AS language-set,
IFNULL(timzone, 'UTC') AS timzone-set
FROM user-settings;

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

20. SELECT record_id,
machine_id,

IFNULL(issue, 'Unknown issue') AS issue_log

IFNULL(technician, 'Not Assigned') AS technician_name
FROM maintenance;

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex